



SARS-CoV-2 Variant ddPCR Assays Instruction Sheet

Description

The Variant SARS-CoV-2 ddPCR Assays are 20X assays designed to detect one of the SARS-CoV-2 Variants in FAM and the Wuhan Strain sequence in HEX using Droplet Digital PCR.

Host

SARS-CoV-2

ddPCR Reagent

One-Step RT-ddPCR Advanced Kit for Probes Cat# 1864021

ddPCR Reaction Setup

Component	Volume per Reaction, μ L	Final Concentration
Supermix	5	1x
Reverse transcriptase	2	20 U/ μ L
300 mM DTT	1	15 mM
20x Assay	1	1x
RNA sample*	5	variable
RNase/DNase-free water	7	–
Total Volume	20**	–

* For positive control, use 1-2 μ L gBlock of stock (see below) ** For the Automated Droplet Generator, prepare 22 μ L per well.

Recommended Thermal Cycling Conditions for Bio-Rad's C1000 Touch Thermal Cycler***

Cycling Step	Temperature, °C	Time	Ramp Rate	Number of Cycles
Reverse transcription	50	60 min	2°C/sec	1
Enzyme activation	95	10 min	2°C/sec	1
Denaturation	95	30 sec	2°C/sec	40
Annealing/extension	55	1 min	2°C/sec	
Enzyme deactivation	98	10 min	2°C/sec	1
Hold (optional)	4	Infinite	2°C/sec	1

*** Use a heated lid set to 105°C and set the sample volume to 40 µL.

Diluting gBlocks (500 nt length)

Make polyA Solution:

1. Poly A stock solution:
 - a. Roche (hydrate with water to 50 µg/µL)-note vortex, it will take a while to dissolve.
 - b. Can also use tRNA as a carrier at the same concentrations.
2. Check solution concentration (50 µg/µL) in TE.
3. Dilute in TE buffer to a final concentration of 100 ng/µL stock
 - a. 100µL poly A into 50 mLs TE

Resuspend gBlocks:

1. Resuspend your lyophilized gblock in 250 µL poly A solution.
 - a. ~200 ng/tube of G-block in 250 µL poly A solution = ~0.8ng/ µL stock solution
2. Vortex to mix and spin down
3. Dilute stock solution with 2-200-fold dilutions for ~48,280 copies/µL working stock (1 µL into 20 µL reaction will produce ~2,400 copies/well)
 - a. 2 µL into 398 µL poly A solution—1:200 dilution = ~4 pg/ µL

4. Repeat
 - a. 2 μ L into 398 μ L poly A solution – 1:200 dilution = ~ 0.02 pg/ μ L
 - b. ~48,280 copies/ μ L working stock $\rightarrow$ ~2400 copies/ μ L QuantSoft at 1 μ L into a well
 - c. Use Poly A solution for any further dilutions desired.
5. Store stock solutions at 4 degrees.